Zoo Network Mainnet Launch: Technical Checklist and Rehearsal Plan

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Contents

1 Overview

This document is the authoritative checklist for Zoo Network mainnet launch. It covers pre-launch validation, rehearsal on staging, launch-day sequence, post-launch operations, and rollback procedures.

Zoo Network is a Layer 2 on Lux mainnet comprising 14 virtual machines (chains) and one Layer 3 subnet (Beluga). All chains share Lux consensus and settle to Lux P-Chain.

Clusters:

- `lux-k8s` — Lux mainnet validators (DOKS, SFO3)
- `lux-test-k8s` — Testnet / staging (DOKS, SFO3)
- `hanzo-k8s` — Shared platform services (IAM, KMS, gateway)

Status legend: `[ ]` = pending, `[x]` = complete, `[!]` = blocked.

2 Pre-Launch: Testnet Validation

All items must be `[x]` before rehearsal begins.

2.1 VM Verification

#	Chain	Criteria	Owner	Status
1	EVM (C-Chain)	Standard EVM conformance tests pass	Core	[ ]
2	DEX (D-Chain)	Order matching, AMM, limit orders verified	Core	[ ]
3	FHE (F-Chain)	Encrypted compute round-trip, TFHE gates correct	Crypto	[ ]
4	I-Chain	DID issuance, revocation, resolution	Auth	[ ]
5	K-Chain	Key management, TLS cert provisioning, rotation	Auth	[ ]
6	T-Chain	MPC wallet creation via FROST DKG	Crypto	[ ]
7	B-Chain	Bridge lock/mint/burn/release full cycle	Bridge	[ ]
8	Z-Chain	ZK proof generation and on-chain verification	Crypto	[ ]
9	Q-Chain	Quantum-resistant signature schemes operational	Crypto	[ ]
10	A-Chain	Agent execution, tool calls, state persistence	AI	[ ]
11	G-Chain	Governance proposals, voting, execution	Gov	[ ]
12	O-Chain	Oracle data feeds, attestation, freshness checks	Data	[ ]
13	R-Chain	Research coordination, dataset registry	Sci	[ ]
14	Beluga (L3)	L3 subnet boots, settles to Zoo L2	Core	[ ]

2.2 GPU Acceleration Benchmarks

#	Workload	Target	Owner	Status
1	CEVM kernel execution (A100)	$\geq$50K tx/s	Core	[ ]
2	TFHE gate evaluation (bootstrapping)	$\leq$12ms per gate	Crypto	[ ]
3	BLS aggregate signature verify	$\leq$2ms / 128 sigs	Crypto	[ ]
4	NTT for ZK proof generation	$\leq$100ms / 2^{24}	Crypto	[ ]

2.3 Testing and Audit

#	Item	Owner	Status
1	E2E Playwright suite: all chains green	QA	[ ]
2	Red team round 1 complete, findings remediated	SecOps	[ ]
3	Red team round 2 complete, findings remediated	SecOps	[ ]
4	Red team round 3 complete, 0 open findings	SecOps	[ ]
5	External audit report received and accepted	SecOps	[ ]
6	Formal proofs: 86 Lean 4 theorems compiled, 0 `sorry`	Formal	[ ]
7	Bridge tested: Lux mainnet ↔ Zoo testnet	Bridge	[ ]
8	Fuzz testing: 72h continuous, 0 crashes	QA	[ ]

3 Rehearsal: Testnet to Staging

Rehearsal runs on `lux-test-k8s`. The full launch-day sequence is executed end-to-end. Any failure resets the rehearsal from step 1.

3.1 Genesis and Bootstrap

#	Step	Owner	Status
1	Generate genesis blocks for all 14 chains	Core	[ ]
2	Validator key ceremony: generate staking TLS certs via K-Chain	Auth	[ ]
3	P-Chain L1 creation transaction submitted and confirmed	Core	[ ]
4	Chain creation transactions for all 14 VMs confirmed	Core	[ ]
5	Validator registration (5 founding nodes) on P-Chain	Core	[ ]
6	MPC wallet creation via T-Chain (FROST DKG for bridge keys)	Crypto	[ ]
7	DID issuance for all 5 validators via I-Chain	Auth	[ ]
8	Verify all 14 chain RPC endpoints responding	Core	[ ]

3.2 Load and Chaos Testing

#	Test	Owner	Status
1	Sustained throughput: 100K+ blocks produced across all chains	Perf	[ ]
2	EVM sustained load: 10K tx/s for 1 hour, no degradation	Perf	[ ]
3	Bridge stress: 1000 concurrent lock/mint operations	Bridge	[ ]
4	Kill 1 of 5 validators, verify chain progress continues	Chaos	[ ]
5	Kill 2 of 5 validators, verify chain halts (BFT threshold)	Chaos	[ ]
6	Restore killed validators, verify chain resumes within 30s	Chaos	[ ]
7	Network partition: split 3/2, verify majority partition progresses	Chaos	[ ]
8	Disk full simulation: verify graceful degradation, no corruption	Chaos	[ ]

3.3 Rehearsal Sign-off

#	Gate	Approver	Status
1	All genesis/bootstrap steps completed without manual intervention	CTO	[ ]
2	Load test targets met	CTO	[ ]
3	Chaos test recovery verified	CTO	[ ]
4	Rehearsal log archived to IPFS	DevOps	[ ]

4 Launch Day

Launch proceeds only after rehearsal sign-off. All times are UTC. A launch coordinator runs the sequence; no ad-hoc changes permitted.

4.1 Genesis and Validators

#	Action	Owner	Status
1	Genesis block signed by founding validators (multisig)	Core	[ ]
2	Bootstrap 5 GPU validator nodes on `lux-k8s`	DevOps	[ ]
3	P-Chain L1 creation confirmed on Lux mainnet	Core	[ ]
4	All 14 chain creation transactions confirmed	Core	[ ]
5	Validator registration confirmed for 5 founding nodes	Core	[ ]

4.2 Service Activation

#	Action	Owner	Status
1	Verify all 14 chain RPCs responding via gateway	Core	[ ]
2	Bridge activation: Lux mainnet ↔ Zoo mainnet	Bridge	[ ]
3	Bridge: first transfer confirmed both directions	Bridge	[ ]
4	Blockscout explorer live at `explore.zoo.ngo`	DevOps	[ ]
5	Explorer indexing all 14 chains	DevOps	[ ]
6	DNS cutover: `api.zoo.ngo` points to mainnet gateway	DevOps	[ ]
7	DNS cutover: `explore.zoo.ngo` points to Blockscout	DevOps	[ ]
8	TLS certificates valid and proxied via Cloudflare	DevOps	[ ]

4.3 Monitoring

#	Action	Owner	Status
1	Prometheus scraping all validator and chain metrics	DevOps	[ ]
2	Grafana dashboards live: consensus, throughput, latency	DevOps	[ ]
3	Alerting rules active: missed blocks, high latency, disk	DevOps	[ ]
4	PagerDuty escalation configured for critical alerts	DevOps	[ ]
5	Structured logging flowing to observability stack	DevOps	[ ]

4.4 Launch Sign-off

#	Gate	Approver	Status
1	All 14 chains producing blocks	CTO	[ ]
2	Bridge operational both directions	CTO	[ ]
3	Explorer indexing confirmed	CTO	[ ]
4	Monitoring and alerting verified	CTO	[ ]
5	Public announcement authorized	CEO	[ ]

5 Post-Launch

5.1 24-Hour Monitoring Window

#	Item	Owner	Status
1	On-call rotation staffed for 24h (2 engineers minimum)	DevOps	[ ]
2	Block production continuous, no gaps >5s	Core	[ ]
3	Bridge transfers monitored, no stuck transactions	Bridge	[ ]
4	Memory and CPU within resource limits on all nodes	DevOps	[ ]
5	No consensus failures or forks detected	Core	[ ]

5.2 Validator Onboarding

#	Item	Owner	Status
1	Community validator documentation published	Docs	[ ]
2	Validator onboarding guide tested by external party	QA	[ ]
3	First community validator registered (target: launch +48h)	Core	[ ]
4	Gradual ramp: 10 validators by week 1, 25 by week 4	Core	[ ]

5.3 Governance and Staking

#	Item	Owner	Status
1	Staking portal activated on `app.zoo.ngo`	Frontend	[ ]
2	Staking contract verified on explorer	Core	[ ]
3	First governance vote: ZIP-001 submitted via G-Chain	Gov	[ ]
4	ZIP-001 voting period completes, result executed on-chain	Gov	[ ]

6 Rollback Plan

If mainnet exhibits critical failures (consensus divergence, bridge exploit, data corruption), the following rollback procedure activates.

6.1 Chain Halt Conditions

A chain halt is triggered if **any** of the following occur:

1. Consensus divergence: two or more validators on different forks for >60s.
2. Bridge exploit: unauthorized mint or release detected.

3. Data corruption: state root mismatch across >1 validator.
4. Critical vulnerability: zero-day with active exploitation.

6.2 Halt Procedure

#	Action	Owner
1	CTO authorizes chain halt	CTO
2	Disable bridge (freeze lock/mint contracts on both sides)	Bridge
3	Coordinate validator shutdown via authenticated channel	DevOps
4	All 5 founding validators stop block production	DevOps
5	Public status page updated: `status.zoo.ngo`	Comms
6	Post-mortem investigation begins (target: root cause within 4h)	Core

6.3 Genesis Reset (Last Resort)

If the chain state is unrecoverable:

1. Export last-known-good state snapshot from all validators.
2. Generate new genesis from snapshot (preserving balances and contract state).
3. Validators re-key via K-Chain key ceremony.
4. Re-execute bootstrap sequence (Section 3.1).
5. Bridge re-activation with fresh MPC keys via T-Chain.
6. Independent audit of new genesis state before restart.

This procedure destroys transaction history after the snapshot. It is used only when all other recovery options are exhausted.

6.4 Communication Plan

#	Action	Owner
1	Status page updated within 5 minutes of halt decision	Comms
2	Community notification via Discord, Twitter, blog	Comms
3	Hourly updates until resolution	Comms
4	Post-mortem published within 72 hours	Core

7 Appendix: Chain Reference

Chain	VM ID	Purpose
EVM (C-Chain)	`cevm`	General-purpose smart contracts (GPU-accelerated)
DEX (D-Chain)	`dex`	Native decentralized exchange with order book
FHE (F-Chain)	`fhe`	Fully homomorphic encryption compute
I-Chain	`identity`	Decentralized identity (W3C DID)
K-Chain	`keychain`	Key management and certificate authority
T-Chain	`thresh`	Threshold signatures and MPC wallets
B-Chain	`bridge`	Cross-chain bridge operations
Z-Chain	`zkvm`	Zero-knowledge proof execution
Q-Chain	`pqc`	Post-quantum cryptographic operations
A-Chain	`agent`	AI agent execution and orchestration
G-Chain	`gov`	Governance proposals and voting
O-Chain	`oracle`	Decentralized oracle network
R-Chain	`research`	Research coordination and datasets
Beluga (L3)	`beluga`	High-throughput L3 subnet